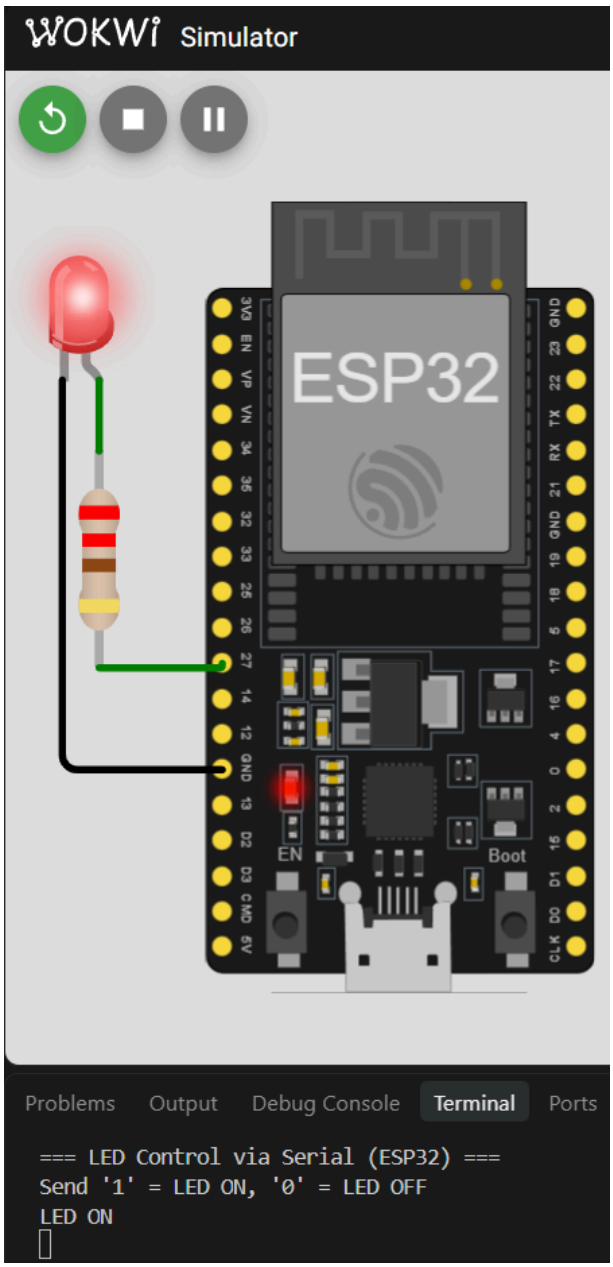
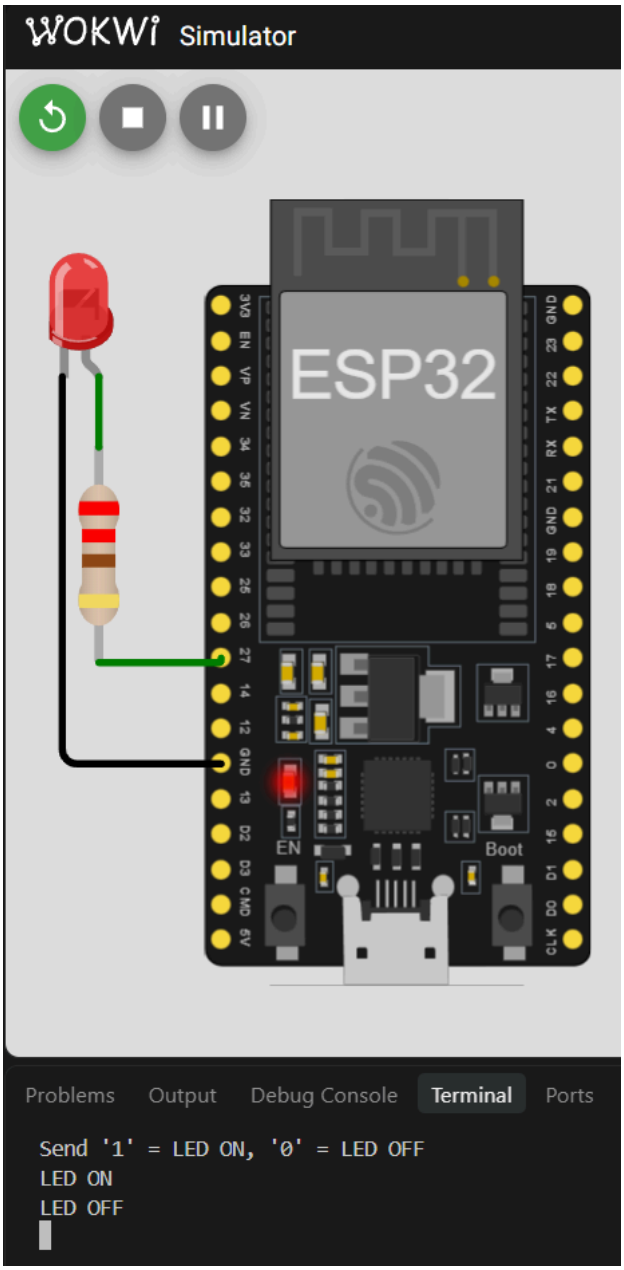
Name: Aaron Ludwig A. Altar **ESP Activity#2**

Section: CPE4B **Date: Sept. 18, 2026**

2.1 LED Control Using Serial Communication

Wokwi Simulation Screenshot

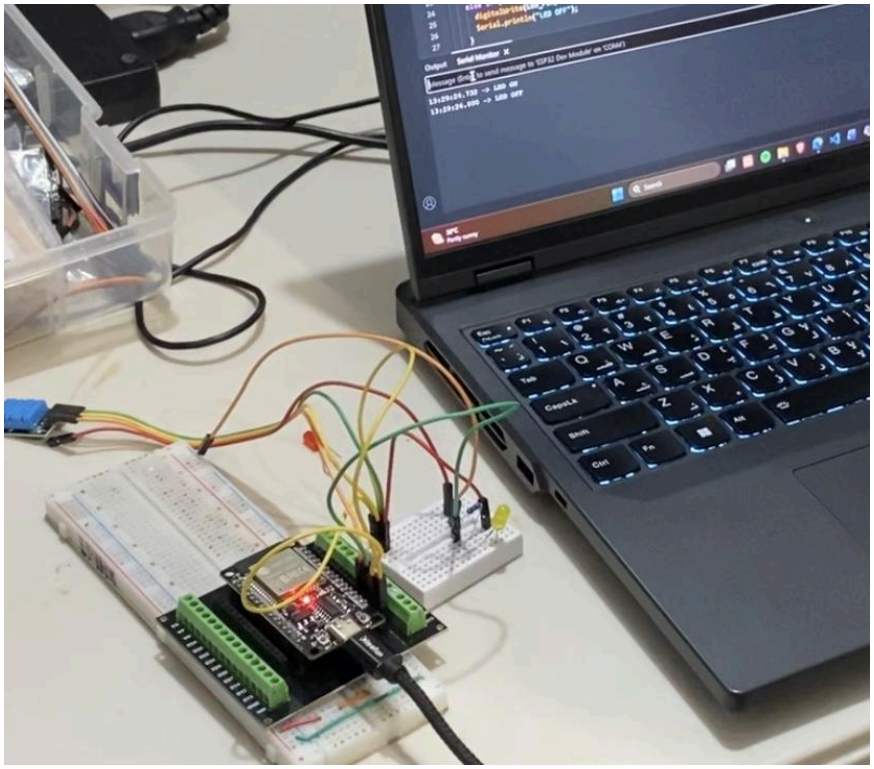


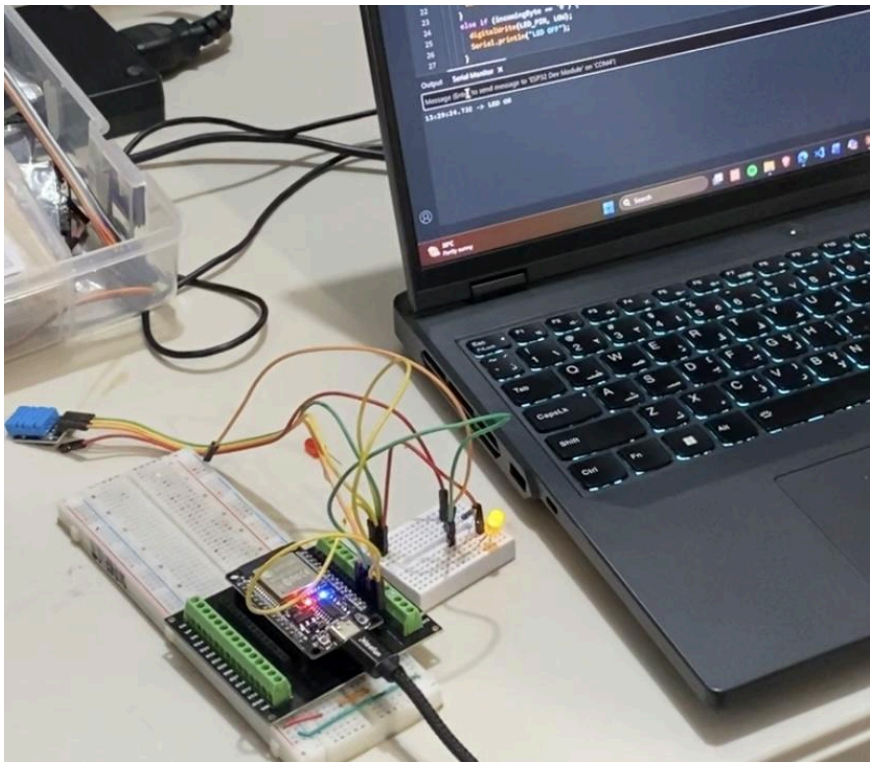


Name: Aaron Ludwig A. Altar **ESP Activity#2**

Section: CPE4B **Date: Sept. 18, 2026**

Actual Experiment





Name: Aaron Ludwig A. Altar **ESP Activity#2**

Section: CPE4B **Date: Sept. 18, 2026**

Code Snippet

```
// EXP 2.1 - Activity #2 (ESP32)
// LED Control Using Serial Communication

const int LED_PIN = 27;  // built-in LED on most ESP32 dev boards
char incomingByte;

void setup() {
  Serial.begin(115200);   // ESP32 default baud rate
  pinMode(LED_PIN, OUTPUT);
  digitalWrite(LED_PIN, LOW);

  Serial.println("=== LED Control via Serial (ESP32) ===");
  Serial.println("Send '1' = LED ON, '0' = LED OFF");
}

void loop() {
  if (Serial.available() > 0) {
    incomingByte = Serial.read();

    if (incomingByte == '1') {
      digitalWrite(LED_PIN, HIGH);
      Serial.println("LED ON");
    }
    else if (incomingByte == '0') {
      digitalWrite(LED_PIN, LOW);
      Serial.println("LED OFF");
    }
  }
}
```

Video Link:

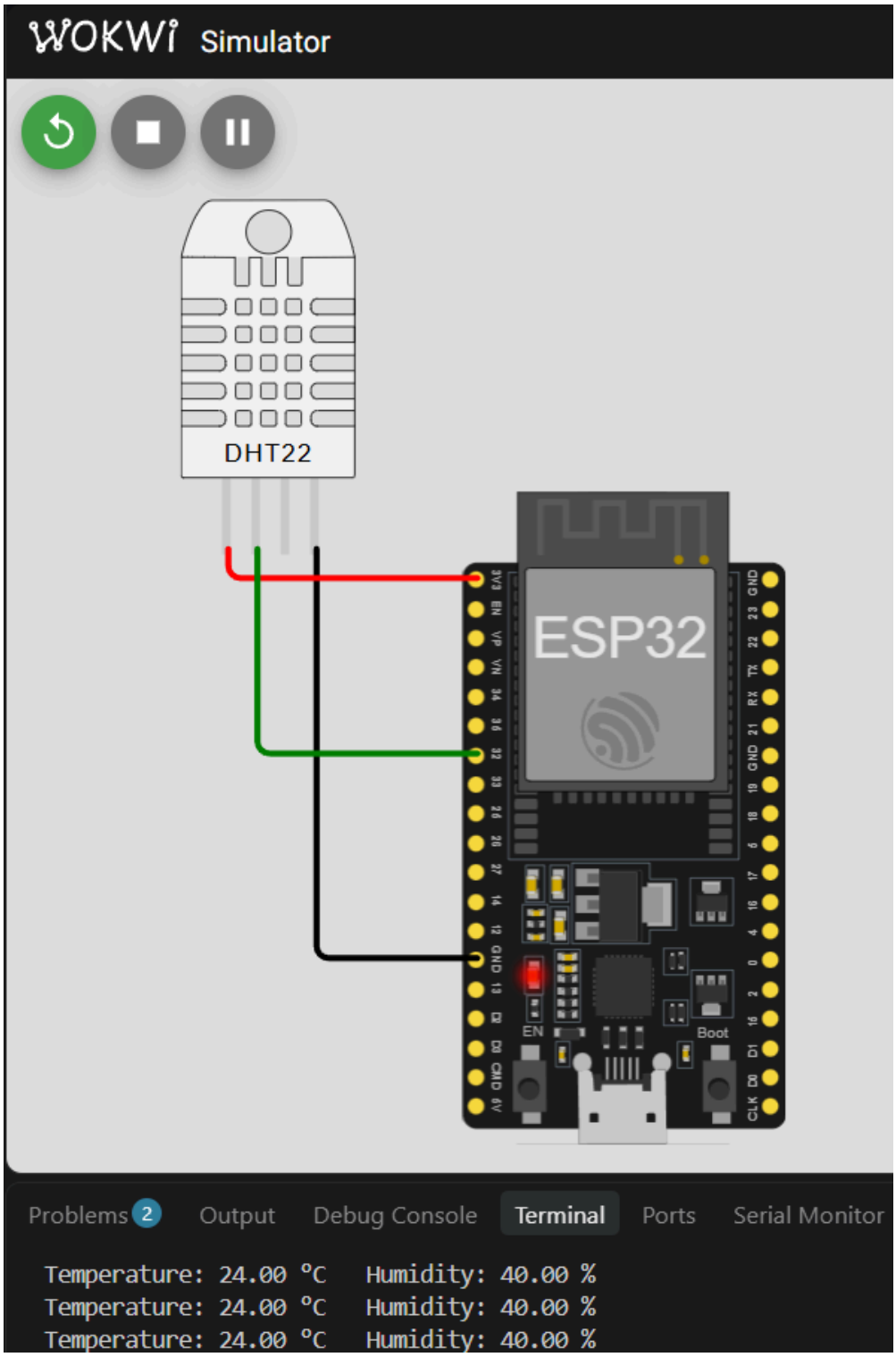
2.1 LED Control Using Serial Communication Video Link

Name: Aaron Ludwig A. Altar **ESP Activity#2**

Section: CPE4B **Date: Sept. 18, 2026**

2.2 DHT11 Monitoring Using Serial Communication

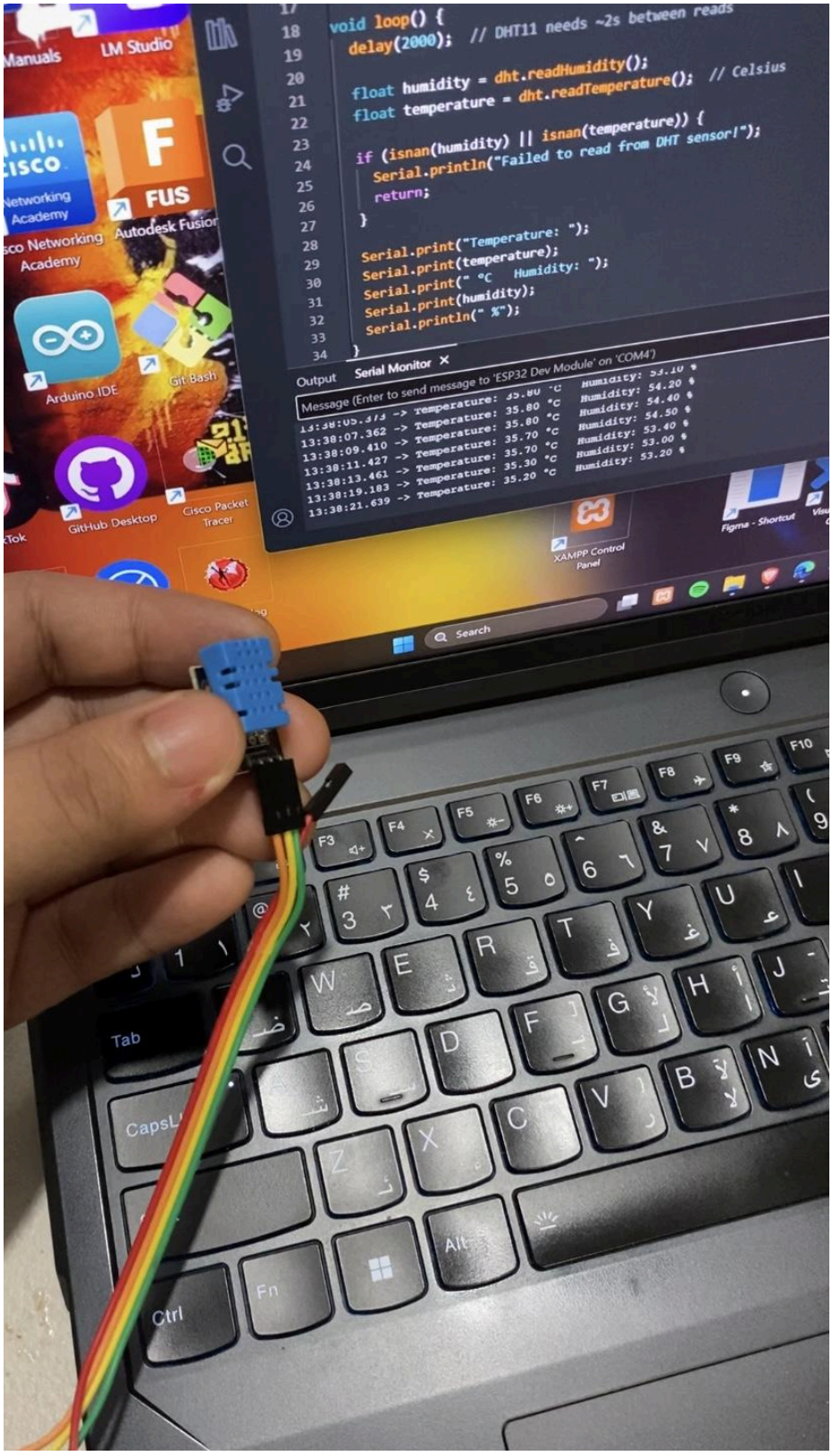
Wokwi Simulation Screenshot



Name: Aaron Ludwig A. Altar **ESP Activity#2**

Section: CPE4B **Date: Sept. 18, 2026**

Actual Experiment



Name: Aaron Ludwig A. Altar **ESP Activity#2**

Section: CPE4B **Date: Sept. 18, 2026**

Code Snippet

```
// EXP 2.2 - Activity #2: Serial Communication
// DHT11 Monitoring via Serial Monitor

#include <DHT.h>
#define WOKWI_SIM

#define DHTPIN 32       // DHT11 data pin connected to pin 32

#ifdef WOKWI_SIM
  #define DHTTYPE DHT22
#else
  #define DHTTYPE DHT11
#endif

DHT dht(DHTPIN, DHTTYPE);

unsigned long previousMillis = 0;
const long interval = 2000;  // read DHT11 every 2 seconds

void setup() {
  Serial.begin(9600);

  dht.begin();

  Serial.println("=== Activity 2: Serial Communication ===");
  Serial.println("DHT11 readings will print automatically...");
  Serial.println();
}

void loop() {
  // ----- Part B: DHT11 Monitoring -----
  unsigned long currentMillis = millis();
  if (currentMillis - previousMillis >= interval) {
    previousMillis = currentMillis;

    float humidity = dht.readHumidity();
    float temperature = dht.readTemperature();  // Celsius
```

Name: Aaron Ludwig A. Altar **ESP Activity#2**

Section: CPE4B **Date: Sept. 18, 2026**

```
    if (isnan(humidity) || isnan(temperature)) {
      Serial.println("Failed to read from DHT sensor!");
      return;
    }

    Serial.print("Temperature: ");
    Serial.print(temperature);
    Serial.print(" °C   Humidity: ");
    Serial.print(humidity);
    Serial.println(" %");
  }
}
```

Video Link

[2.2 DHT11 Monitoring Using Serial Communication Video Link]